Which brain cells carry the genetic risk for myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS), and is the pattern specific to the disease?

Research abstract

Background. Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a debilitating illness affecting an estimated 0.8 to 3.3 million people in the United States. While the 2025 DecodeME genetic study (15,579 cases and 259,909 controls) linked ME/CFS risk to brain tissue, tissue-level analyses cannot distinguish among hundreds of resident cell types. This study sought to identify the specific brain cell populations that carry ME/CFS genetic risk compared to other brain traits.

Methods. Using a pre-registered protocol established prior to analysis, genetic signals were mapped to genes and evaluated across 461 cell clusters from the Human Brain Cell Atlas. A specificity score compared ME/CFS enrichment against a panel of 19 comparison traits. Methodological validity was supported by verifying known biology in control traits, and findings were tested through permutation testing, locus-drop checks, and secondary analyses.

Results. Eleven cell clusters, primarily excitatory neurons in the cortex and amygdala, were significantly enriched for ME/CFS genetic risk. Six of these clusters were more specific to ME/CFS than to comparison traits. The strongest specific signal mapped to an excitatory cluster from amygdala dissections, designated Amex_153, which remained specific across every change made to the comparison panel. The same clusters also carried signal in a large chronic-pain cohort. Within ME/CFS, the signal was clearest in patients whose illness began with an infection.

Conclusions. These results point to excitatory neurons in cortical and limbic circuits as cell populations where ME/CFS genetic risk is concentrated. Because independent datasets did not confirm the specific amygdala anatomical label, these findings offer a hypothesis regarding cortical and limbic excitatory neurons for future cohorts to test. To our knowledge, no prior study has tested whether ME/CFS cell-type enrichment is specific against a panel of other traits.